

Version 13.1 of Unicon for Microsoft Windows

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Abstract

This report describes how to install and run the Unicon programming language under the Microsoft Windows environment. It is the primary reference for the Windows-specific language extensions to Unicon.

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1 1. Introduction

The Microsoft Windows implementation of Version 13.1 of Unicon runs on computers with a 32-bit or 64-bit Windows operating system. It has been tested primarily on Windows 10 and Windows 7, with some limited testing on older versions such as Windows XP.

This implementation of Unicon is an open source project under the GNU license; it may be copied and used with few restrictions. The Unicon Project makes no warranties of any kind as to the correctness of this material or its suitability for any application. The responsibility for the use of Unicon lies entirely with the user.

The basic reference for Unicon is the book *Programming with Unicon* [1]. This book is available from the Unicon Project web site at unicon.org. The graphics facilities are detailed in a separate report [2]. Unicon is an open source product based on the volunteer work of many people; final responsibility for this release rests with Clinton Jeffery of University of Idaho and Jafar Al Gharaibeh of Architecture Technology Corporation. Send requests, and bug reports to jeffery@cs.uidaho.edu. General Unicon language questions can be sent to unicon-group@lists.sourceforge.net, which is a moderated low-volume mailing list. Only list members can post messages to the list.

2 2. Installing Windows Unicon

Windows Unicon is distributed as a single executable installer file 14+ MB in size. Two sets of the binaries are available, 32-bit and 64-bit, from sourceforge.net/projects/unicon/files/. 32-bit binaries might run on 64-bit Windows, but 64-bit Windows binaries will not run on 32-bit Windows. Once the binaries are downloaded, run the installer (setup-unicon*.exe) which guides you through the process of setting up Unicon for Windows.

During installation, you will be asked for a drive and directory into which the Unicon files will be installed, which defaults to C:\UNICON. Files will be installed into several subdirectories of the location you specify. Installation also results in the creation of a Windows Unicon program group with a set of icons that allow you read on-line documentation, uninstall the software, and launch Windows Unicon.

Within the directory that Unicon was installed, the most important sub-

directory is the `bin` directory, which contains the Unicon system binaries. Several executable binary files are required to use Unicon, including the executor and the compiler that produces virtual machine code files for the executor. The executor and compiler can be invoked from an integrated development environment called *Ui*, a software tool that provides a visual interface to the programming process. Applications that provide a graphical interface are usually constructed by means of a visual interface builder called IVIB, a drawing program that generates Unicon code for a program's user interface. The following table summarizes the contents of the `bin` directory:

unicon.exe Unicon VM compiler

icont.exe Icon VM compiler

iconx.exe Icon VM executor

ivib.exe Improved Visual Interface Builder, constructs graphical user interfaces

ui.exe Unicon IDE (integrated development environment)

wunicon.exe, wicont.exe, wiconx.exe Win32 versions, used internally by *Ui*, do not require a console

Important! Unicon's `bin` directory, normally `C:\UNICON\BIN` usually must be added to your `PATH` specification. This is normally done by the installer program, but it can also be done through the Control Panel, or on the command line, when necessary.

3 3. Running Windows Unicon

Windows Unicon may be used by means of the programming environment, *Ui*. You may also invoke the compiler and run programs directly from the command-line. This section provides a brief discussion of the command-line tools. Instructions on using the programming environment are given in [UTR 12](#).

Command line execution of Unicon programs is similar on many platforms and is described in the Unicon Programming Language book and other reports. Windows Unicon includes a command-line compiler (`unicon`) and

virtual-machine interpreter (iconx) that run from the command-line. For example, an Unicon program in the file prog.icn is compiled into virtual machine "icode" by the command:

```
unicon prog.icn
```

The extension .icn is optional on the command line, since the compiler assumes the .icn extension by default. For example, it is sufficient to use

```
unicon prog
```

Windows Unicon also includes a compiler (wicont) and virtual-machine interpreter (wiconx) that can be invoked from the command-line, but are usually invoked by Ui. These executables, unlike the command-line versions, do not require a console window (cmd.exe), and thus are useful for creating programs like Ui that are launched from a shortcut or menu. The instructions below refer to icont and iconx, but generally would also be true for wicont and wiconx. The Windows icode files that result of compilation have the extension .exe, for example, the above compilation would produce a file prog.exe. The program may then be executed as:

```
prog
```

Alternatively, unicon can be instructed to execute the icode file after translation by appending a -x to the command line, as in

```
unicon prog.icn -x
```

If unicon is run with the -x option, the file prog.exe is left and can be run subsequently using an explicitly named executor as described above.

4. Testing the Installation

There are a few programs included in the distribution that can be used for testing the installation and getting a feel for running Unicon. They live in a subdirectory tests\\samples under the Unicon installation directory.

hello.icn This program prints the Unicon version number, time, and date. To run this test, launch the Ui program, open file hello.icn, and select the Make and Run menu options. To test the command line tools:

```
unicon hello
hello
```

Note that this can be done in one step with

```
unicon hello -x
```

roman.icn This program converts Arabic numerals to Roman numerals. To run this test, launch the Ui program, open file roman.icn, and select the Make and Run menu options. To test the command line tools:

```
unicon roman -x
```

and provide some Arabic numbers from your console.

If these tests work, your installation is probably correct and you should have a running version of Windows Unicon.

5 5. More on Running Unicon

For simple applications, the instructions for running Unicon given in Section 3 may be adequate. The Unicon compiler supports a variety of options that may be useful in special situations. There also are several aspects of execution that can be controlled with environment variables. These are listed here. If you are new to Unicon, you may wish to skip this section on the first reading but come back to it if you find the need for more control over the translation and execution of Unicon programs.

5.0.1 5.1 Arguments

Arguments can be passed to the Unicon program by entering them in Ui's Arguments... item in the Run menu, or appending them to the command line. Such arguments are passed to the main procedure as a list of strings. For example,

```
prog text.dat log.dat
```

runs the icode file prog.exe, passing its main procedure a list of two strings, "text.dat" and "log.dat". The program also can be translated and run with these arguments with a single command line by putting the arguments after the -x:

```
unicorn prog -x text.dat log.dat
```

These arguments might be the names of files. For example, the main procedure might begin as follows:

```
procedure main(args)
  in := open(args[1]) | stop("cannot open file")
  out := open(args[2], "w") | stop("cannot open file")
  .
  .
  .
```

5.0.2 5.2 The Compiler

The Unicorn compiler can accept several Unicorn source files at one time. When several files are given, they are translated and combined into a single icode file whose name is derived from the name of the first file. For example,

```
unicorn prog1 prog2
```

translates the files prog1.icn and prog2.icn and produces one icode file, prog1.exe. In addition to supplying files on the command line, files may be linked or included using appropriate commands in the source file.

A name other than the default one for the icode file produced by unicorn can be specified by using the -o option, followed by the desired name. For example,

```
unicorn -o probe prog
```

produces the icode file named prog1.icode rather than prog1.exe.

If the -c option is given to unicon, the translator stops before producing an icode file and an intermediate “ucode” file with the extension .u is left for future use (normally they are deleted). For example,

```
unicon -c prog1
```

leaves prog1.u, instead of producing prog1.exe. These ucode files can be used in a subsequent unicon command by using the .u name. This saves translation time subsequently. For example,

```
unicon prog2 prog1.u
```

translates prog2.icode and combines the result with the ucode files from a previous translation of prog1.icode. Ucode files also can be added to a program using the link declaration.

The informative messages from the translator can be suppressed by using the -s option. Normally, both informative messages and error messages are sent to standard error output.

The -t option causes &trace to have an initial value of -1 when the icode file is executed. Normally, &trace has an initial value of 0.

The option -u causes warning messages to be issued for undeclared identifiers in the program.

5.0.3 5.3 Environment Variables

When an icode file is executed, several environment variables are examined to determine execution parameters. The values assigned to most of these variables should be numbers.

Environment variables are particularly useful in adjusting Unicon’s storage requirements. Particular care should be taken when changing default values: unreasonable values may cause Unicon to malfunction.

The following environment variables can be set to adjust Unicon’s execution parameters. Their default values are listed in parentheses after the environment variable name. Percentages listed are percentages of the physical main memory available on the system.

BLKSIZE (1%) This variable determines the size, in bytes, of the initial region in which Unicon allocates lists, tables, and other objects. If additional block regions are needed, they may be smaller.

COEXPSIZE (.01%) This variable determines the size, in words, of each co-expression block.

ICONFONT (fixed) This variable contains the name of the default font used by Windows that are opened. Any fixed-pitch font may be the default. Example syntax is:

::

set ICONFONT=Lucida Sans Typewriter

IPATH (undefined) This variable contains a list of directories, separated by semicolons. The directories are searched for library files specified by link declarations, as well as imported packages and inherited classes. Unicon system directories ipl/lib, uni/lib, plugins/lib, uni/gui, and uni/3d are added onto the end of any user-supplied IPATH.

LPATH (undefined) This variable contains a list of directories, separated by semicolons. The directories are searched for header files specified by preprocessor `$include` directives. Unicon system directories ipl/incl, uni/lib, plugins/lib, uni/gui, ipl/gincl, and ipl/mincl are added onto the end of any user-supplied LPATH.

MSTKSIZE (0.25%) This variable determines the size, in words, of the main interpreter stack.

NOERRBUF (undefined) If this variable is set, `&errout` is not buffered.

STRSIZE (1%) This variable determines the size, in bytes, of the initial region in which strings are stored. If additional string regions are needed, they may be smaller.

TRACE (undefined) This variable initializes the value of `&trace`. If this variable has a value, it overrides the translation-time `-t` option.

WICONLOG (undefined) This variable contains the name of a file to which output from `wicont` and `wiconx` will be written after their execution completes.

6 6. Features of Windows Unicon

Windows Unicon supports all the features of prior versions of Unicon and Version 9.4 of Icon, including graphics facilities, with the following exceptions and additions:

- Path specifications can be entered using either a / or a \\. Examples are:

A:	ICON	TEST.ICON
A:/ICON/TEST.ICON		

- Wildcards are allowed in filename command-line arguments.
- wunicon and wicont allow an option -q that causes it to exit on termination without waiting for a user keystroke; this is useful in batch compilation.
- system() supports launching Windows applications from inside Unicon.
- A console window opens if any of the (non-redirected) standard input, output, or error files is accessed.
- .bmp image files are supported for reading and writing.

6.0.1 6.1 Native Windows User Interface Access

Version 13 of Windows Unicon provides the following built-in user interface components that make use of native Windows features. In some cases, the facilities described here are integrated into the Unicon gui classes so that they are used automatically in Unicon programs that provide a visual interface. This section summarizes features; the following section provides a reference guide.

The goal of the native facilities is not to provide the entire Windows repertoire, any more than the entire X Window repertoire is provided to UNIX users. Instead, features have been chosen that are (1) important to the Windows look and feel, (2) general enough to be implementable on other platforms, and (3) can co-exist or be integrated with existing IPL facilities. These facilities, and the "Wu" IDE that uses and demonstrates them, may be more accessibility-friendly than Unicon's normal GUI interface.

Menus A menu bar is created with a call like:

```
WinMenuBar(W, ["&File", "&Open", "&Save", "E&xit"],  
            ["&Edit", "C&ut", "&Paste", "C&opy"],  
            ["&Help", "&About"])
```

This function converts approximately the top text line of the window into a menu bar. The appearance of the above example is given in Figure

1. When menu items are selected, they are produced as entire strings

(such as "&Open") by Event().

| | Figure 5: a Windows menu bar

Scroll Bars A scroll bar is created with a call like

WinScrollBar(W, "sb_1", x, y, wd, ht) This function places a scrollbar with a particular size and position, which default to a standard size on the right edge of the window. The appearance of a typical scroll bar is illustrated in Figure 6. When scroll bar activity takes place, the scroll bar's string id is produced (in this case, "sb_1") by Event(), and &x and &y are both set to the scroll bar's position.

| | Figure 6: a Windows scroll bar

Buttons A button is created with a call like

WinButton(W, "hello", x, y, wd, ht) This function places a button with a particular size and position. The size defaults to a standard size large enough display the button's label. The appearance of a pair of buttons is illustrated in Figure 7. When a button is pressed, the button's string label is produced (in this case, "hello") by Event().

| | Figure 7: a pair of Windows buttons

Common Dialogs Several common dialogs are provided for selecting colors, fonts, and files to open or save. These functions return an attribute value or a file name. Examples are illustrated in Figures 8-11.

| | Figure 8: the Windows color dialog

| | Figure 9: the Windows font dialog

| | Figure 10: the Windows open dialog

| | Figure 11: the Windows save dialog

6.0.2 6.2 Native Windows Facilities Function Reference

The following functions provide a small but useful subset of the native Windows interface. They are useful mainly for applications that require a native look and feel, and since they are nonportable, they are deprecated. Graphical interfaces are best created for Unicon applications using the Improved Visual Interface Builder and its GUI class library.

fattrib(x, s) - get file attribute This Windows Icon function is subsumed by `stat()` and is no longer built-in. You can add the library declaration "link fattrib" to your program if you don't want to change your `fattrib()` calls.

WinAssociate(s) - return the application associated with an extension This Windows Icon function produces the name of the program associated with a string file extension supplied in parameter `s`.

WinButton(s, x, y, wd, ht) - install button

| **WinButton(s, x, y, wd, ht) installs a pushbutton on w with label s.**

Whenever the button is clicked, the string label `s` is placed on the window's event queue.

\ ht height of text in system font * 7/4

WinColorDialog(s) - select color `WinColorDialog(s)` executes the Windows common dialog for color selection with default color `s`. Returns a string attribute value corresponding to user's selection, or fails if the user selects Cancel. This function is called from the Unicon Program Library procedure of the same name.

WinEditRegion(s, s2, x, y, wd, ht) - manipulate edit region `WinEditRegion(s, s2, x, y, wd, ht)` manipulates a Windows edit region named `s`. This flexible editor is limited to text that is < 32Kbytes in length. The operation

performed depends on argument s2. If argument s2 is omitted, WinEditRegion(s) returns a string containing the current contents of region s. If s2 is supplied and does not start with a !, it is a string to be edited; lines are separated by \\r\\n. s2 strings starting with ! are commands:

- WinEditRegion(s, "!clear") clears the current selection..
- WinEditRegion(s, "!copy") copies the current selection.
- WinEditRegion(s, "!cut") cuts the current selection.
- WinEditRegion(s, "!paste") pastes the current selection.
- WinEditRegion(s, "!modified") succeeds if region s has been modified since it was last assigned a value
- WinEditRegion(s, "!setsel") sets the selection using parameters x and y as indices..
- WinEditRegion(s, "!undo") undoes the previous operation, if possible.

WinFontDialog(s) - Execute Windows common dialogs for font selection, with default font s. Returns a string attribute value corresponding to user's selection, or fails if the user selects Cancel.

WinMenuBar(w, L1, L2, ...) - install a menu bar on w. Each list, which presently may contain only strings, describes one popup menu on the menu bar. The first element of the list is the menu's name that appears on the bar. The remaining elements are the menu options for that popup menu. After a menu bar is installed, Event() returns the strings. For example, after a call: WinMenuBar(["File", "Open", "Save"], ["Help", "About"]) the menu bar would show File and Help menus, and anytime the user selected the Open, Save, or About menu options, "Open", "Save", or "About" would be queued on the event list. Ampersand is used to specify keyboard shortcuts, as in ["&File", "&Save", "Save &As..."].

WinOpenDialog(s1,s2,i,s3,j,s4), WinSaveDialog(s1,s2,i,s3,j,s4) - Execute Windows common dialogs for opening and saving files. These functions open typical dialogs to perform file selection for reading and writing, respectively. s1, s2, and i are the caption, default value, and text entry field size. s3 and j specify the filter string and its index. s3 is a string of alternating names and filters, separated and ending in \, of the form "name1|filter1|name2|filter2\...\\". It defaults to "All Files(\.\\)\.\\\". j supplies the default extension index within s3; it defaults to first pair in filter string. s4 is the directory to show when the dialog is opened; it defaults to use Windows version-specific rules. These functions return the file name chosen. They fail if the user selects Cancel.

WinPlayMedia(w, s1, s2, ...) - **play a multimedia resource.** String arguments ending in .wav are presumed to be wave sound file names. Strings ending in .mid or .rmi are presumed to be MIDI files. All other strings are treated as MCI command strings, and processed by the Windows Media Control Interface.

7 7. Known Bugs, Limitations, and Problem Areas

Windows Unicon has a number of limitations inherent to the platform, as well as some remaining bugs that may get fixed.

General

- Pipes are not well-supported. Some versions of Windows are more functional than others, but in general, mode "p" file I/O should not be relied upon.
- For files opened in the translated mode, the position produced by seek() may not reflect the actual number of characters written because of the translation of carriage-return/line-feed sequences to line-feed characters.

- `system()` cannot launch DOS or NT console applications. Such applications can often be run by launching a command-line interpreter to execute them, as in

```
system("cmd /C /Q " || command)
```

- `system()` may return before the launched application completes if that application opens no window, closes its main window before its execution is finished, or if several windows are opened shortly after the application is launched.
- Under Windows 95, the focus does not return to the command line window after `wicont` or an Unicon program executed by `wiconx` is launched. Redirecting standard input without also redirecting standard output can cause spurious runtime errors. Redirecting standard input from an empty file is not detected; it is treated as no redirection and a console window is opened for input if the standard input file is read.

Graphics Facilities

- Attribute `linestyle` is ignored by Windows when the `linewidth` is greater than 1; linewidths greater than 1 are always drawn using a solid line style.
- Attribute `fillstyle` does not correctly interpret the value masked. When masked fills are requested, textured fills are performed instead.
- In addition to the standard values "dashed", "striped", "solid", attribute `linestyle` has values "longdashed", "dashdotted", and "dash-dotdotted" corresponding to Windows' standard pen styles for longer dashes and alternating longer and shorter (Icon-standard size) dashes in lines.
- Mutable colors do not work unless the window is the active window, and sometimes do not work as expected.
- Colors are never freed.

- Windows may use dithered colors, resulting in an unattractive appearance in applications where solid colors are expected. Most colors are dithered on (now vanishingly rare, but still in use) 16-color machines (not 16-bit color: 16-color), and color-intensive applications are ugly or unusable on those systems.
- References to attributes `display`, `iconpos`, `iconimage`, `iconlabel`, and `visual` always fail, as do calls to function `Default()`.
- Attribute pointer supports only "arrow", "cross", "ibeam", "uparrow", and "wait", depicted by the following images. The appearance of these pointers varies slightly on different versions of Windows.
- Although some program examples using attribute `drawop` work, `drawop` does not work the same way as on other platforms.

8 8. Reporting Problems...or Successes

Problems with Windows Unicon should be noted on a trouble report form (TROUBLE.TXT, included with the distribution) and sent to

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In order to minimize response times, the preferred method for reporting problems is by e-mail. I would also like to hear suggestions and success stories from satisfied users; e-mail is great but letters and postcards are even better for this kind of feedback. Think of TROUBLE.TXT as a "registration" for your free software.

8.0.1 Acknowledgements

Unicon is a superset of and is based on the Icon programming language, whose design and implementation was supported, in part, by grants from the National Science Foundation. Many people contributed to the Icon (and therefore Unicon) implementations. Most of the credit for coordinating Icon's implementation goes to Ralph Griswold. Gregg Townsend has made extensive contributions and currently maintains Icon.

Much of this work was made possible by support from the National Library of Medicine Specialized Information Services division, under the direction of Phillip Thomas.

9 References

1. Clinton Jeffery, Shamim Mohamed, Jafar Al Gharaibeh, Ray Pereda, and Robert Parlett, Programming with Unicon, 2nd edition, <http://unicon.org/ub/ub.pdf>, 2016.
1. Gregg M. Townsend, Ralph E. Griswold, and Clinton L. Jeffery, Graphics Facilities for the Unicon Programming Language Version 9.3, The Univ. of Arizona Icon Project Document IPD281, 1996.

















